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Bioinformatics for Beginners

Projects

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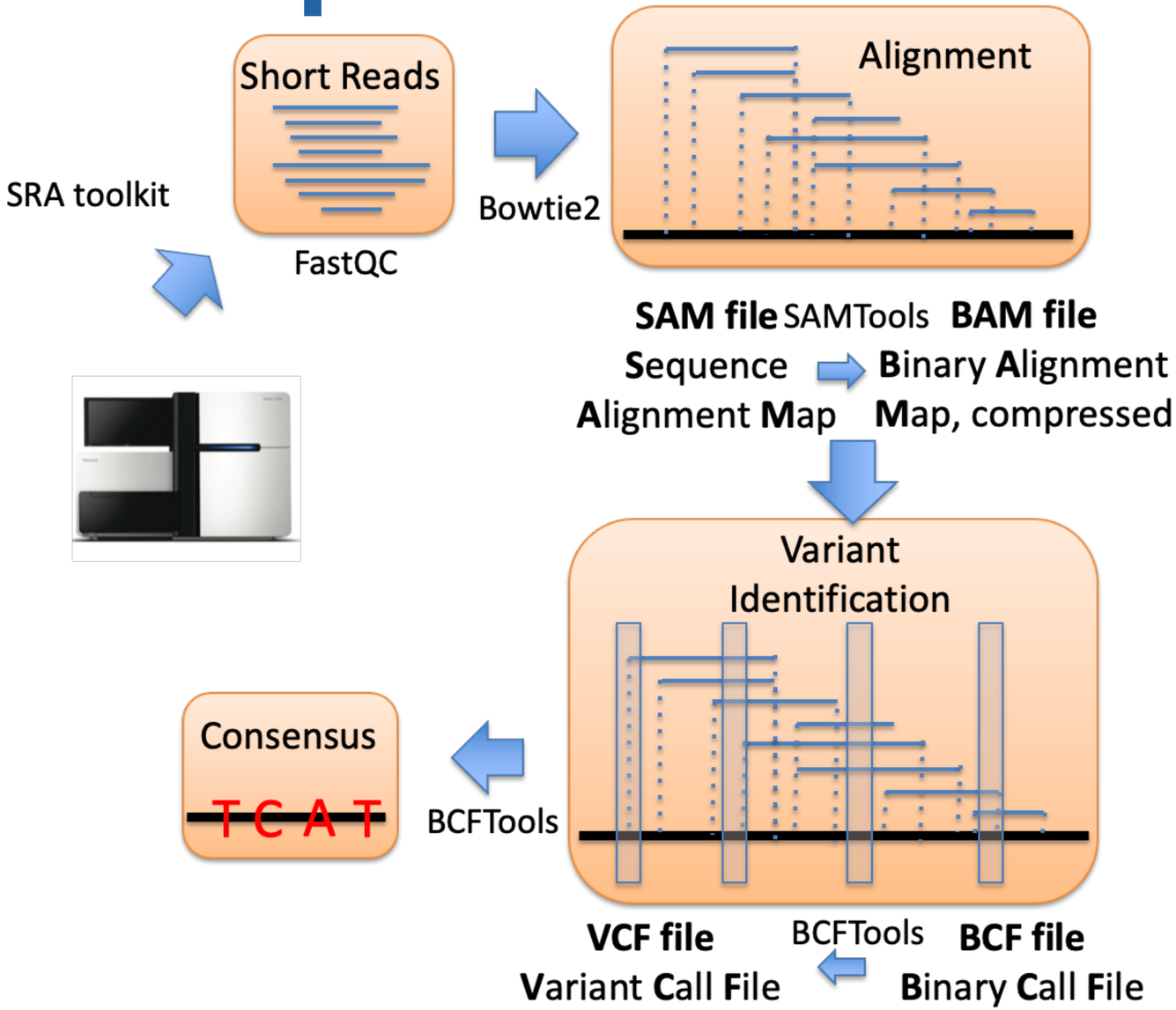
What we have done so far?

Genomics Pipeline



GitPod- an online platform
Earth Cluster
your local machine

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Smart-Seq pipeline

Smart-seq2 is a popular FACS based protocol for single-cell RNA-sequencing. Therefore it is not that high through put. But you can choose which cells you want to sequence, even the rare ones since you have knowledge and therefore control over cell types. With these protocols, you have information on the full transcriptome read. Reads are aligned to the appropriate reference genome using STAR. Smart-Seq2 has the highest sensitivity, higher capture ability and lower cost compared to the droplet based 10X.

Today we have the Github Tutorial:

TASK 2: Run STARSolo to generate a gene expression count matrix

Comparative Analysis of Single-Cell RNA Sequencing Methods

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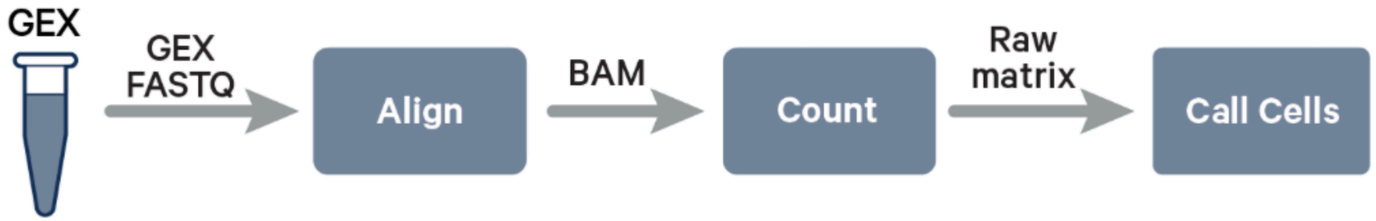
10X Pipeline

10x genomics has developed several protocols for single-cell RNA-sequencing. Unlike Smart-Seq, 10x protocols are droplet-based, and use unique molecular identifiers (UMIs) to avoid counting an RNA fragment more than once. 10X is highest through put in the market, therefore also expensive. It does not give full sequence information. Typically, 10x data contains significantly more cells, sequenced at lower depth, compared to smart-seq. Alignment and feature quantification are done with 10x cellranger software.



10X Tools: Cell Ranger and Space Ranger

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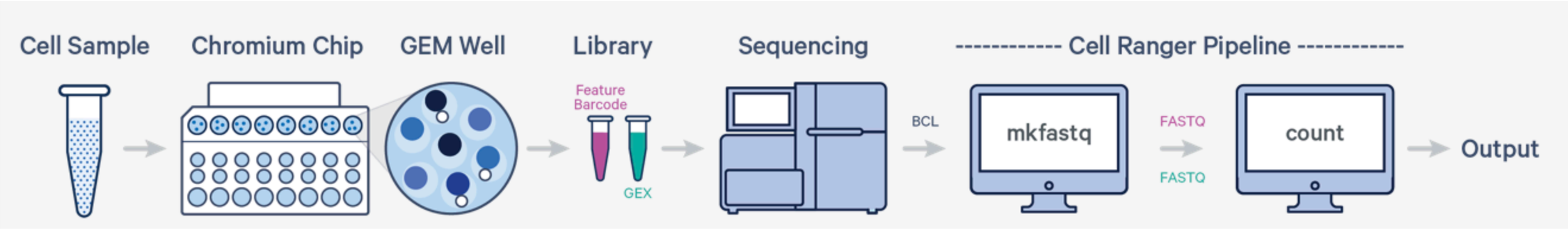
Visium Spatial Gene Expression is a next-generation molecular profiling solution for classifying tissue based on total mRNA.

Cell Ranger is a set of analysis pipelines that process single cell data to align reads, generate feature-barcode matrices, perform clustering etc.

Space Ranger is a set of analysis pipelines that process Visium Spatial Gene Expression data. Space Ranger allows users to map the whole transcriptome in formalin fixed paraffin embedded (FFPE) and fresh frozen tissues to discover novel insights into normal development, disease pathology, and clinical translational research.



Single Cell RNA Sequencing Analysis

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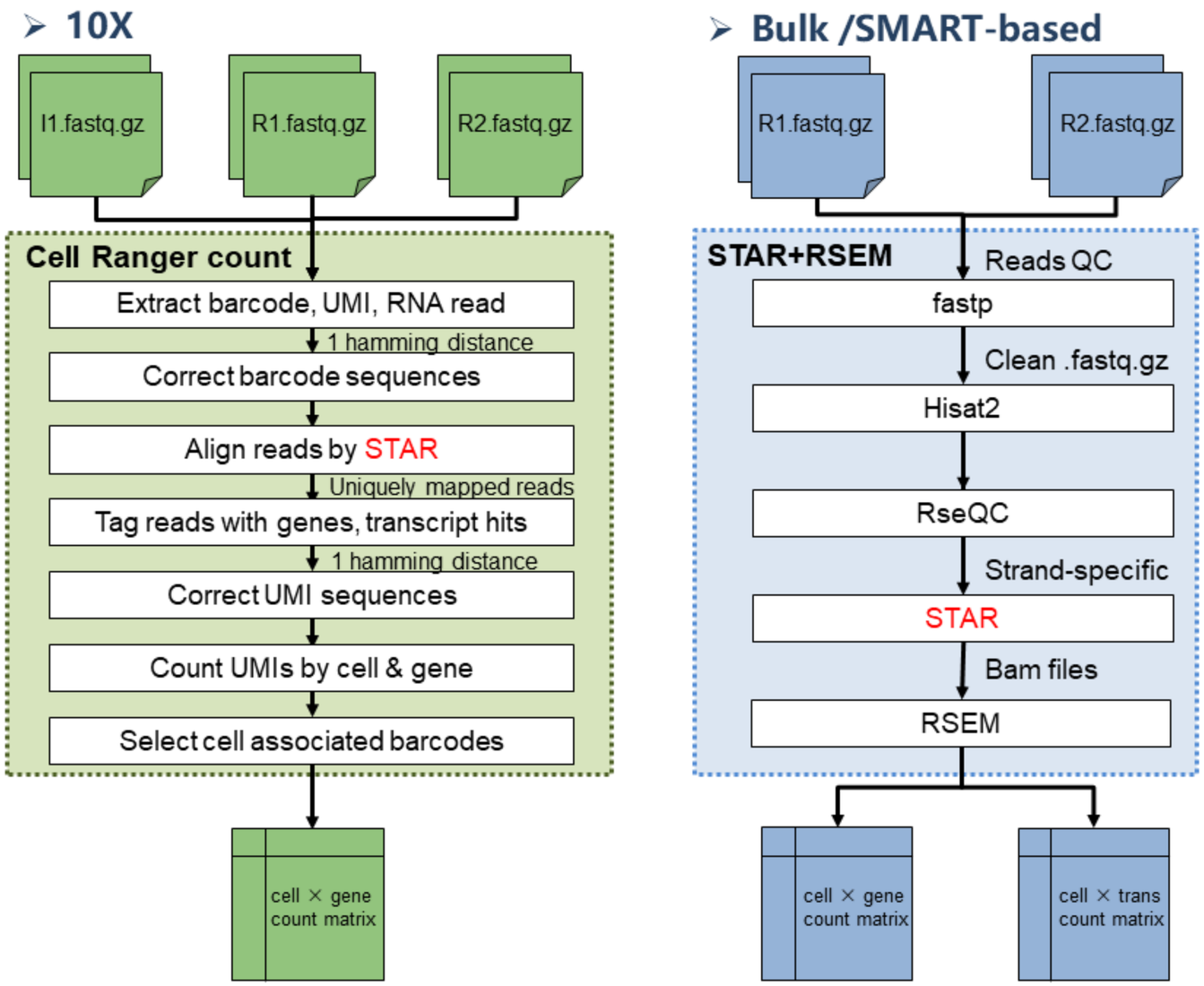

Single Cell RNA Sequencing Analysis

Quality Control → Normalization → Dimension Reduction → Clustering

Low Read Cells
Empty Barcodes
Double Cell Barcodes(Doublets)

PCA(Principal Component Analysis)
t-SNE(t-distributed stochastic neighbor embedding)

Clustering Algorithm

R Analysis
-Seurat Package

Phython Analysis
-Scvi and Scanpy

https://satijalab.org/

https://www.helmholtz-munich.de/en/icb/research-groups/theis-lab

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What can you do with these?

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Downstream Analysis

- **Variant Analysis:** Based on Variant Call Files
  - check out mutations and their consequences with Variant Effect Predictor
  - study mutated genes and their functions with Gene Ontology Analyses
- **Phylogenetics:** Based on Variant Call Files create trees based on the identified mutations, can be based on comparisons across family or population
- **Characterization:**
  - clustering analyses to identify groups of cells with similar profiles. Do this either for expression or mutation profiles
  - cell type determination with or without spatial data
- **Gene Expression Analysis:** determine differentially regulated genes and study their functions

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Possible Directions for Genomics Projects

- identify single nucleotide variants by redoing the genomics pipeline for more samples either to do variant analysis, or create trees at the family or population level
- identify copy number variations in tumors
- genotype short tandem repeat mutations to study their impact
- study mutations to cluster cells

Possible Directions for Transcriptomics Projects

- identify differentially regulated genes and check out their functions
- cluster cells based on their profiles (images, cell type, disease, location etc.)
- For both either start from the alignment step or skip, either work with Smart-seq or 10X, either do with or without spatial data

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Projects

- We will offer projects but feel free bring in your own ideas, datasets etc. Simply ask us.
- Select something today, and start working on it in the last hour.
- You have two weeks to work on it and on your presentation
- Ideally pick sth you can finish. Challenge yourself. Empower yourself.
- You may work as a group or individually
- You may work on the same project independently with different data sets etc.
- Presentation: We expect 3-4 min of talk per participant, you may describe your question, methods, results, your discussion briefly, showing some slides

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Project Selection

https://docs.google.com/spreadsheets/d/1b6qGHuwVzpD7inAoKyiWcC3fH9ZDMUHmjaPs6C05gy0/edit?usp=sharing

Please enter your project in this list once we introduce our proposed projects. Confirm your topic with Tugce AND your project leader if exists